Pre-Registered Falsification Targets for Bubble Spacetime Theory

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Purpose

This document pre-registers specific, falsifiable predictions of Bubble Spacetime Theory (BST) before experimental results are available. BST derives physical constants from the geometry D_IV^5 = SO_0(5,2)/[SO(5) x SO(2)] using five integers (rank=2, N_c=3, n_C=5, C_2=6, g=7, N_max=137) and zero free parameters.

We commit to the following predictions. If any is contradicted by measurement at the stated precision, BST is falsified at tree level.

Prediction 1: BaTiO3 137-Plane Casimir Peak

Claim: A Casimir force measurement in a BaTiO3 wedge cavity will show an anomalous peak when the gap equals exactly 137 lattice planes (54.9 nm for the a-axis, 137 x 4.009 A).

Falsification condition: No anomalous feature within +/- 5 planes of the 137-plane spacing, at measurement sensitivity sufficient to resolve 5% modulation of the Casimir force.

Estimated cost: $25,000 (wedge cavity + PFM measurement).

BST basis: N_max = 137 is the spectral cap of D_IV^5. Boundary conditions at 137 lattice planes resonate with the complete eigenvalue spectrum.

Prediction 2: No Neutrinoless Double Beta Decay at Any Scale

Claim: BST predicts Dirac neutrinos (Z_3 on CP^2 fiber, Hopf structure forces Dirac). Neutrinoless double beta decay (0vbb) will NOT be observed at any half-life, regardless of detector sensitivity.

Falsification condition: Observation of 0vbb at any half-life above 10^24 years (current sensitivity ~10^26 years, next generation targets 10^28 years).

Timeline: LEGEND-200 (~2027), nEXO (~2030), CUPID (~2028).

BST basis: T319 (permanent alphabet), Hopf fiber argument — neutrino mass structure is Dirac, Majorana mass term topologically forbidden by D_IV^5 fiber structure.

Prediction 3: W Boson Mass = 80.361 GeV

Claim: m_W = n_C * m_p / (8*alpha) = 80.361 GeV.

Falsification condition: Combined 2026 LHC measurement gives m_W outside the range [80.35, 80.37] GeV at 3-sigma confidence.

BST basis: Electroweak sector of D_IV^5. See BST_W_Boson_Precommitment.md.

Prediction 4: Three Generations Exactly

Claim: N_c = 3 forces exactly three fermion generations. No fourth-generation fermion exists at any mass.

Falsification condition: Discovery of a fourth-generation charged lepton or quark at any collider energy.

BST basis: Z_3 symmetry of color sector is exhaustive. N_c = 3 is a topological invariant, not a parameter.

Prediction 5: CMB Spectral Index

Claim: n_s = 1 - n_C/N_max = 1 - 5/137 = 0.9635.

Falsification condition: Planck/LiteBIRD combined measurement gives n_s outside [0.960, 0.967] at 3-sigma.

Current status: Planck 2018 gives n_s = 0.9649 +/- 0.0042. BST prediction is 0.3-sigma from central value.

BST basis: Cascade fingerprint of D_IV^5 on CMB (Toy 1401).

Standing Commitment

All predictions above are parameter-free. None can be adjusted after the fact. If any single prediction is falsified at the stated precision, the corresponding BST derivation chain is wrong and must be publicly retracted.

Casey Koons. Bubble Spacetime Theory. [DATE] Repository: github.com/[repo]. Prior art: Zenodo DOI 10.5281/zenodo.19454185 (Working Paper v20, April 7, 2026).